

Standard Operating Procedure

Startup Procedure for Crude Distillation Unit Atmospheric Column

Document No: BPL-SOP-CDU-003 | Rev: 02

1. Purpose

To define the safe and structured startup procedure for the CDU Atmospheric Column C-101 following a planned turnaround or emergency shutdown.

2. Safety Precautions

- Ensure all blinding is removed and signed off.
- Verify H2S detectors are functional.
- PPE Required: Fire Retardant Coveralls, Hard Hat, Safety Goggles, Gas Monitor.

3. Step-by-Step Procedure

- Nitrogen Purging: Establish N2 flow to C-101. Monitor O2 levels at overhead vent until $O_2 < 2\%$.
- Steam Out: Introduce low-pressure steam (LP) to the bottom of C-101 to displace N2 and heat the column.
- Circulation: Start crude circulation pump P-101A/B. Establish cold circulation through the desalter and heat exchanger train.
- Furnace F-101 Light Up: Ignite pilot burners on F-101. Slowly increase fuel gas flow. Ramp up temperature at 15°C/hr .
- Column Heating: As crude outlet temp reaches 250°C , monitor level in C-101 bottom. Start bottom pump P-102 when level reaches 40%.